Joshtalks Growth Assessment

About Joshtalks : Voice of India

Voice of India is an evaluation platform that measures how well AI systems work under Indian linguistic, cultural and deployment conditions. It was built by AI4Bharat at IIT Madras and Josh Talks, and launched in New Delhi on 4 August 2026. Our mission is to make AI understand India as it actually speaks, and to measure honestly how close it is.

What sets us apart :
• Real speech, real conditions. Results are broken down by geography, audio quality, speaking rate, gender, device and age. District-level error rates in our benchmark range from about 4% to 44%.
• Honest numbers. Our paper finds that no evaluated system consistently reaches a 20% word error rate across all 15 languages, and that public benchmarks can overstate real-world performance.
• Research behind the boards. Methods are published, not just posted.• Data nobody else has. A private, held-out test set across India's regional clusters.

In simple words, we care about questions like:

- Which speech system actually works best for a given real use?
- Where does a system fail: people talking over each other, noisy rooms, accents, phone lines?
- Can a published accuracy number be trusted and compared across companies?
- Is the human-labelled data behind a model or a benchmark actually correct?

Many systems look impressive in demos. Real quality only shows when they are tested properly.

Why evaluation matters

A good evaluation is not only a score. It is a system that helps people make better decisions. Good evaluations help teams:

- compare systems fairly
- find failure cases
- understand what users actually need
- decide what to build, buy or fix

Why this challenge exists

This challenge tests how you think when the problem is not fully defined. We are intentionally not giving you exact steps or a fixed answer. We want to see whether you can:

- research a field you may not know yet
- make clear choices and explain them
- work carefully with real-looking data
- turn findings into rules a team can act on

How this assessment works

- You have **7 days** from when you receive this to submit everything.
- There is no page limit. Write as much as you need, but keep it easy to read; bullets and tables are welcome.

- Question 1 is research and design. Question 2 is data analysis.
- For Question 2 you also get a data workbook and RTTM files. The links are inside Question 2.

What we are evaluating

- judgment and the reasons behind your choices
- depth of research
- attention to detail
- quality of execution
- product thinking

Question 1:

Design a speaker diarization evaluation

What speaker diarization is

Speaker diarization answers "who spoke when" in a recording. A system listens to a conversation and returns segments, each with a start time, an end time and a speaker label such as Speaker 1 or Speaker 2. It is not told who the speakers are or how many there are. It is used in meeting notes, call centre analytics, medical scribes and subtitles.

What makes it hard: people talking over each other, very short turns ("mm-hm"), many speakers, similar voices, background noise, far-away microphones and phone-line audio.

What a good diarization eval needs

- A fixed test set that looks like real use
- A reference: the correct answer, usually built and annotated by people
- A metric that turns the gap between a system's output and the reference into a number
- The same conditions for every system
- A way to say how sure you are that one system really beats another

Learn more: datasets, benchmarks and papers

Read a few of these to see how real diarization evaluations are built. You don't need to copy any of them.

Name	What it is	Link
AMI Meeting Corpus	100 hours of recorded meetings with detailed annotations	groups.inf.ed.ac.uk/ami/corpus
ICSI Meeting Corpus	About 70 hours of natural meeting recordings	groups.inf.ed.ac.uk/ami/icsi
CALLHOME American English	Unscripted two-person telephone calls; CALLHOME telephone speech is a classic diarization test set	catalog.ldc.upenn.edu/LDC97S42
DIHARD III	Diarization challenge on hard audio from varied real-world domains	dihardchallenge.github.io/dihard3
VoxConverse	Multi-speaker clips from online videos, with diarization labels	github.com/joonson/voxconverse
AliMeeting	Mandarin meetings recorded with both far-field and close-talking microphones	openslr.org/119

Name	What it is	Link
CHiME-8 Task 1 (DASR)	Meeting transcription and diarization across different devices, rooms and speaker counts	chimechallenge.org, CHiME-8 Task 1
The Third DIHARD Diarization Challenge (Ryant et al.)	Paper describing how DIHARD III was designed and scored	arxiv.org/abs/2012.01477
A Review of Speaker Diarization (Park et al.)	Survey of diarization methods, datasets and metrics	arxiv.org/abs/2101.09624
dscore	Open-source scoring tool for diarization (DER and JER)	github.com/nryant/dscore
pyannote.metrics	Toolkit for evaluating and analysing diarization errors	pyannote.github.io/pyannote-metrics
MeetEval	Toolkit for scoring transcripts with speaker labels (cpWER, tcpWER)	github.com/fgnt/meeteval

The challenge

Josh Talks is launching a public leaderboard that ranks speaker diarization systems. Companies building meeting tools, call analytics and voice products will use it to choose a system. Design the evaluation methodology from scratch. You don't have to follow any existing benchmark.

What your main document must cover

1. Purpose: who the leaderboard is for, and what decision each user makes with it
2. Models
   - The types of systems you would evaluate, for example open-source models you run yourself, commercial APIs, speech-to-text services that also label speakers, and systems that work live
   - A named list of at least 10 specific models or services, with the company, the exact model name, the type and how you would run it (self-hosted or API, you don't need to know all the technical aspects)
   - How you keep the comparison fair between such different systems
3. Test data
   - What conversations you would record or collect: speakers, settings, languages or accents, devices, and roughly how many hours
   - How the reference (the correct answer) is built, and how you would check its quality
4. Metrics
   - Your primary metric and supporting metrics, each defined in plain words (a formula is welcome)

- Your scoring settings and why: for example the collar (a tolerance around boundaries), whether overlapping speech is scored, and whether systems are told the number of speakers

5. Evaluation process
   - Step by step, from raw audio to a published leaderboard
   - How you decide that one system really beats another, rather than winning by chance
   - What you publish next to each score

Optional, and a strong plus: run a small version

Pick 2 or 3 short public recordings (for example from VoxConverse or AMI), run 2 or 3 systems on them (open source or free tiers), score them with dscore or pyannote.metrics, and report what you saw. Write down exactly which system, version and settings you used.

Deliverables for Question 1

1. Main document: everything above, ending with the final reflection below.
2. One-page report: the eval, the core setup, your most important design choices and the main takeaway, written so a busy reviewer understands it in two minutes.
3. Leaderboard sketch: how the results would look as a product, for example the main table, the filters and what sits next to each score. A simple mockup, slides, Figma or a clear hand sketch is fine.
4. Video walkthrough (compulsory, 2-3 minutes at most): what you designed, the 2 or 3 choices that matter most and why, and what you learned from the benchmarks you read.

Final reflection

End your main document with short answers to these questions:

1. Which design choice are you least sure about, and why?
2. Which benchmark you read influenced you most, and how?
3. What could make your leaderboard misleading, and how did you guard against it?
4. What would you improve with more time?

End of Question 1. Question 2 below is a separate task. It has no connection to your Question 1 answer, and you do not need your Question 1 design to solve it.

Question 2:

Diagnosing Deepgram Nova-3 Performance on Voice Arena Diarization-Bench

Context:

Question 2 is a separate task from Question 1. In Question 1 you design a diarization evaluation from scratch. In Question 2 that work has already been done: Voice Arena has built its own speaker diarization benchmark, run the leading systems on it and published the results.

You are not asked to design or rerun the benchmark. You are asked to investigate one result on it: why Deepgram Nova-3 scores so badly on some recordings and well on others.

1. Start with VoiceArena leaderboard

Leaderboard: voicearena.com/diarization-bench

Spend some time on it before you open the data. It shows how every system scored, how each score splits into the three error types (missed speech, false alarm and speaker confusion), and how results change when you filter by setting, number of speakers and amount of overlapping speech. Note where Nova-3 sits and which error type drives its score.

2. How the benchmark was run

- **Test set:** 139 real conversations recorded by Voice Arena, about 22 hours of scored speech. The recordings and their correct answers are held out and not published as a dataset.
- **Settings:** in-person conversations, where everyone in the room is recorded on one shared room microphone, and online calls, where each person's audio is mixed into one file.
- **Speakers:** 2 to 8 people per conversation, speaking English with US, European, Indian and Vietnamese accents.
- **Systems:** commercial APIs and open-source models. Every system receives the exact same audio file, and every output is scored in exactly the same way.
- **Human reference (the correct answer):** who spoke when, marked by trained annotators who listened to each speaker's own close microphone, and checked by a second annotator.
- **Main metric:** diarization error rate (DER), the share of speech time a system gets wrong. Lower is better.
- **Strict scoring:** no tolerance window around the start and end of each turn, overlapping speech is scored, and every vocal sound counts, including short fillers and laughter.

3. What you get for Question 2

- **A data workbook:** 15 of the 139 recordings, with audio links, scores, word counts, transcripts, audio measurements and Deepgram re-runs with different settings.

- **RTTM files:** the human reference, the Nova-3 output and a voice activity detector output as timed segments, so you can recompute the scores yourself.
- **The Deepgram playground:** so you can run Nova-3 on the same audio and check our numbers.

More details about the data are given in the sheets. Each tab explains what its columns mean, and the Read me tab defines every term used in the pack. Read it first.

4. What you need to do

- **Explore:** understand where Nova-3 stands on the leaderboard and what kind of error it makes.
- **Investigate:** use the data pack and your own playground runs to find out why Nova-3 fails on some recordings and not on others, and whether the cause is on Deepgram's side or ours.
- **Communicate:** write it up as a report and social posts (Parts A, B and C below).

You do not need a speech or machine learning background. If something in the task is unclear, state your assumption in your report and carry on.

Why this challenge exists

On our diarization benchmark of 139 recorded conversations, Deepgram Nova-3 has a diarization error rate (DER) of 67.1%, far behind the other systems we test. Most of that error is missed speech.
We want a clear, fair and evidence-backed explanation of what is happening. That is your job.

How it works

- **Time:** 7 days from the day you receive this pack.
- **Tools:** anything you like, including spreadsheets, code and AI assistants.
- **Background:** no speech or machine learning background is needed. The data pack explains every term. We care about how you reason from evidence.

The data

You get a workbook, **Deepgram_Failure_Analysis_Data.xlsx**, covering 15 real recordings from the benchmark. They are not labelled (where nova-3 performs well and where it performs bad); working out which is which is part of the job.

Tab	What is in it
Read me	What the file contains, how to run Nova-3 yourself, and plain definitions of every term
Sessions	One row per recording: audio link, setting, date, recording station, accent, speakers, length, overlap
Results	Nova-3 DER and its parts, speakers found, words returned and speech covered, next to the DER of six other systems

Tab	What is in it
Transcripts	The human transcript for each person next to what Nova-3 and ElevenLabs Scribe v2 returned for that person
Deepgram API check	What the API reported back for 6 recordings: file length, channels, model, diarizer, warnings, confidence
Audio quality	File format, loudness, clipping, background noise and speech-above-noise for each file
Deepgram settings	The poor recordings re-run with different Deepgram languages and models

You also get **Deepgram_Failure_RTTM_Files.zip**: the human reference, the Nova-3 output and the VAD output as timed speaker segments (RTTM files), one per recording, so you can recompute the scores yourself. A README inside explains the format.

Do not take our numbers on trust. Open the Deepgram playground, paste audio links from the Sessions tab and run them. New Deepgram accounts get free credit.

- **At least 5 recordings:** 3 that score badly and 2 that score well, with the benchmark settings (model nova-3, diarization on, utterances on).
- **At least 2 other settings** on one of the poor recordings, for example a different language option or a different model.
- **Proof of every run:** a screenshot or screen recording that shows the audio link, the settings and the output. Include the request ID if the playground shows one.

Compare what you get with our results. If your output differs from ours, say so and explain why that might be.

Learn more

Short reads to understand more about the benchmark, evaluation metrics and deepgram model.

- Deepgram: speaker diarization
- Deepgram: multilingual code switching (language=multi)
- pyannote.metrics, the toolkit behind our DER scores (Interspeech 2017)
- Voice Arena Diarization-Bench leaderboard

The challenge

Part A: Find out why Nova-3 fails

Use the data to answer these questions. Back every claim with numbers from the pack.

1. **What exactly goes wrong?** Describe the failure in measurable terms. Which part of the error dominates, and how does it differ from the recordings where Nova-3 does well?

2. **Is the problem Deepgram's output, or our side of the test?** Check the human reference, the VAD, the audio files, the settings we used, and the way we turn Deepgram's words into speaker timelines. Say for each one whether it could explain the failure, and show why.
3. **What can you rule out?** List the explanations you tested and rejected, and the evidence that rejects each one.
4. **What is the root cause?** Give your best explanation, how confident you are, and what you would test next to confirm it. Be clear about what the data can and cannot prove.
5. **What should change?** Recommendations for Deepgram, and anything Voice Arena should change in how it runs or reports the benchmark.

Part B: Write the report

Turn your analysis into a report that we can publish on the Voice Arena blog and send by email to Deepgram's research team. Write it to that standard: the strongest submissions may be adapted for both. It has two readers at once: speech researchers who will check every number, and a general technical audience who wants the story.
Name the report as : **Deepgram Nova-3 Performance on Voice Arena Diarization-Bench**

Structure it like a short research note:

- **Summary:** four to six bullet points that state the finding and the evidence.
- **How to read your charts:** a few lines so a newcomer can follow them.
- **Overview table:** one row per recording you analysed, with the numbers that matter most.
- **Evidence per recording:** for at least 4 poor and 2 good recordings, a chart or playground screenshot with two or three bullet points on what it shows.
- **What you ruled out, and why.**
- **Root cause:** your best explanation, how confident you are, and what would confirm it.
- **What this means:** recommendations for Deepgram and for Voice Arena, and the limits of your analysis.

Keep the tone factual and fair. The aim is to help Deepgram fix a problem, not to embarrass them.

Part C: Write the social posts

Write the posts Voice Arena would publish from its official accounts when the report goes live:

- **X:** one post or a thread of up to 5 posts.
- **LinkedIn:** one post of up to 1,300 characters.

Both must be accurate to your report, readable by someone who has never heard of DER, and something a careful company would be comfortable posting about another company's model. Use [link] where the report link would go.

What to submit

	Deliverable	Format
1	The report (Part A answered inside it, plus Part B)	PDF, no page limit
2	The X post or thread and the LinkedIn post (Part C)	One document
3	Proof of your Deepgram playground runs (screenshots or screen recording)	Images, PDF or a video link
4	Your working: the spreadsheet, notebook or code that produced your numbers and charts	Any format we can open
5	One short 2-3 min video explaining your findings	Video link

Name your files **FirstName_LastName_Report.pdf**, **FirstName_LastName_Posts**, **FirstName_LastName_Proof**.

What we are evaluating

- **Diagnosis:** do you find what actually drives the failure, and is it supported by the data?
- **Reproduction:** did you run Nova-3 yourself, and does your proof match what you claim?
- **Attention to detail:** are your numbers right, consistent ?
- **Communication:** could the report go to Deepgram's researchers and our blog readers as it is?
- **Judgement:** are the posts accurate, clear and fair to the company whose model is being discussed?

Final reflection

End your report with a short note: what surprised you in the data, and what you would investigate within one more week.

Materials :

Deepgram_Failure_Analysis_Data.xlsx *: Deepgram_Failure_Analysis_Data*
RTTM files : *Deepgram_Failure_RTTM_Files*
This pack contains real recordings and real model outputs. Please use them only for this assessment and do not share them.